

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Brainstorming & Idea Prioritisation

ShopEZ — MERN Stack E-commerce Application

Phase: Brainstorming & Ideation Phase

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Classification: Project Documentation — For Academic Submission

Brainstorming & Idea Prioritisation

1. Introduction

This document records the structured brainstorming and ideation process undertaken at the inception of the ShopEZ project. The objective of this phase was to collaboratively identify user pain points, generate candidate solutions, and systematically prioritise features using established frameworks. The output directly informs the Problem Statement Definition and Requirement Analysis phases.

2. Problem Statement Selection

Small and medium-sized businesses lack the technical resources to establish a customised, scalable online storefront, while end-consumers demand a seamless, fast, and secure shopping experience — encompassing intuitive search, dynamic filtering, persistent cart state, and trusted multi-gateway payment processing.

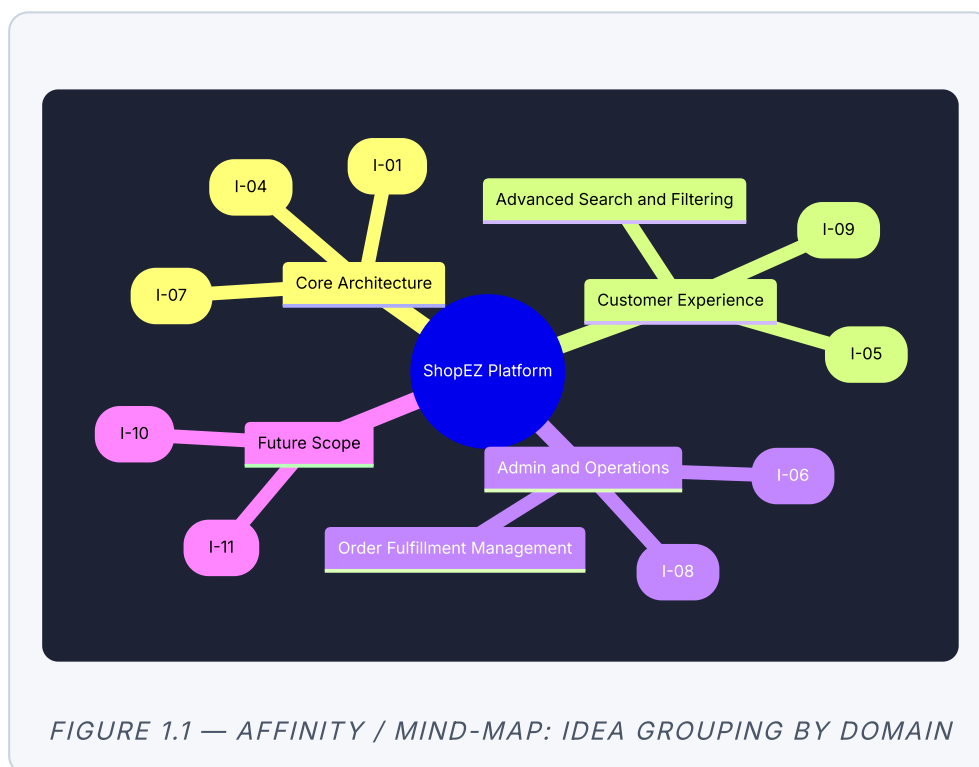
3. Idea Generation & Grouping

3.1 Raw Idea Listing

Idea ID	Idea Description	Category	Retained?
I-01	Build a custom MERN stack application for full architectural control	Core Architecture	✓ Yes
I-02	Use Shopify as a SaaS platform	Existing Platforms	✗ Discarded — insufficient customisability
I-03	Use WooCommerce (WordPress)	Existing Platforms	✗ Discarded — plugin overhead, no SPA support
I-04	Implement JWT-based stateless authentication with HTTP-only cookies	Security	✓ Yes
I-05	Integrate a Payment Gateway (Stripe / Razorpay)	Payments	✓ Yes
I-06	Use Cloudinary CDN for optimised media/image storage	Media Management	✓ Yes
I-07	Implement Role-Based Access Control (Admin vs. Customer)	Access Control	✓ Yes
I-08	Build an interactive Admin Dashboard with real-time sales metrics	Admin Operations	✓ Yes

Idea ID	Idea Description	Category	Retained?
I-09	Persist shopping cart state using MongoDB for cross-device continuity	User Experience	✓ Yes
I-10	Implement AI-driven product recommendations	Advanced Features	● Future Scope
I-11	Multi-vendor support	Marketplace Features	● Future Scope

3.2 Idea Grouping — Affinity Diagram



4. Idea Prioritisation — MoSCoW Framework

Priority	Classification	Feature / Idea	Rationale
Must Have	Core — Release 1	User Registration & JWT Authentication	Foundational to all user-specific operations
Must Have	Core — Release 1	Product Catalogue (CRUD Operations)	Primary business function; no storefront without it
Must Have	Core — Release 1	Shopping Cart (Add, Update, Remove)	Essential to purchase flow
Must Have	Core — Release 1	Multi-Step Checkout Flow	Core revenue- generating pathway
Must Have	Core — Release 1	Payment Gateway Integration	Order completion requires payment confirmation
Should Have	Release 1	Admin Dashboard (Sales Charts)	Enables business decision-making
Should Have	Release 1	Product Reviews & Ratings	Builds consumer trust and aids discovery
Should Have	Release 1	Cloudinary Image Management	Improves performance; required for scalable image serving

Priority	Classification	Feature / Idea	Rationale
Could Have	Release 2	Wishlist Functionality	Improves retention, not blocking to launch
Could Have	Release 2	AI Product Recommendations	Non-trivial to implement; deferred to future roadmap

5. Conclusion

The brainstorming phase successfully distilled the project vision into a focused set of prioritised deliverables. The **Must Have** features constitute the minimum viable product (MVP) scope for Sprint 1 and Sprint 2. All accepted ideas map directly to User Stories defined in the *Requirement Analysis* phase.